

HelixVPN — Master Glossary (every term, acronym, protocol, RFC)

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Revision: 2 **Last modified:** 2026-06-26T12:00:00Z **Status:** active — Volume 0 (Spine, meta & governance) nano-detail document **Rev 2:** Added challenges (vasic-digital/challenges, §11.4.27) + containers (vasic-digital/containers, §11.4.76/.161) submodule entries; added a HelixError entry cross-referencing its R7 rename to the canonical CoreError. **Authority:** Subordinate to [../SPECIFICATION.md](#) §11 (the spine glossary, which this document expands). Where this glossary disagrees with the spine on a term's meaning, the spine wins until amended per §11.4.73.

Document role. The single, comprehensive, alphabetical glossary for the whole HelixVPN final/ specification set: every domain term, acronym, protocol, RFC, crate, service, and product name used across Volumes 0–10. Each entry is a precise 1–3-sentence definition plus the **owning doc** where the term is specified in depth. It expands the 20-row spine glossary ([SPEC §11]) to full coverage. Where a term names a fact not yet proven on hardware (a measured number, an interop result), the dependency is marked UNVERIFIED (§11.4.6) and the owning doc carries the gate that resolves it. No term is invented; every entry traces to a source doc or RFC.

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How to read an entry

Each entry is **Term** — **definition. (owning doc / source)**. Owing-doc shorthand mirrors [SPEC §10] and [MASTER_INDEX]:

- **SPEC** = SPECIFICATION.md; **00...11** = the topical/WBS overview docs; **99** = source-coverage ledger.
 - **v02-.../v03-.../v04-.../v05-.../v06-.../v10-...** = the nano-detail volume documents.
 - Citation ids [04_ARCH \$N], [04_P0/P1/P2], [04_UI], [05_YB0], [SYN] are the source-research ids defined in [SPEC] metadata.
 - A trailing UNVERIFIED marks a term whose *quantitative* claim awaits a named gate (G1–G6) or benchmark; the **term** is real, the **number** is the gate's to prove.
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Numerals & symbols

- **1→N (one user → N networks)** — HelixVPN's headline differentiator (X1): a single user is given policy-scoped access to *many* joined private networks at once, with per-user ACL routing, rather than one VPN per network. (00 §4.2; v02-data-plane/routing-and-addressing.md; FR-301)
 - **4-layer test coverage** — the §11.4.4(b) floor: pre-build gate + post-build + runtime/on-device + paired §1.1 meta-test mutation, applied to every fix and feature. (10; §11.4.4)
 - **4via6** — Tailscale-style scheme that maps an advertised IPv4 LAN into the tenant's IPv6 ULA space, so two networks both exposing e.g. 192.168.1.0/24 never collide; the overlapping RFC1918 range is encoded as a host suffix inside a /96 derived from the tenant /48. The D4 Camp-A answer. (01; v02-data-plane/routing-and-addressing.md; v03-control-plane/svc-ipam.md; D4; FR-303)
 - **8-platform reach** — the portability differentiator (X5): iOS, Android, Linux, macOS, Windows, Web (Console-scoped), HarmonyOS NEXT, Aurora OS — all from one shared Rust core + Flutter UI. (03; v04-client/*; HVPN-NFR-600..609)
 - **8-criteria MVP DoD** — the Phase-1 Definition-of-Done: self-host-from-zero, enroll connector+client, authorized-reach + denied-unauthorized, auto-escalate-to-MASQUE, policy-edit <1s, revoke <1s, kill-switch+DNS-leak, no-durable-log + all-three-apps-drive-it. (00 §10.2; 07; FR §M)
 - **100.64.0.0/10 (CGNAT space)** — the shared-address (carrier-grade NAT) range proposed by the D4 Camp-B sources (GMI/KMI) as a 1:1-per-network alternative to 4via6; surfaced, not chosen for MVP. (D4; 99; v03-control-plane/svc-ipam.md)
 - **§1.1 (paired mutation)** — the constitution's meta-test discipline: every gate ships a deliberate mutation that must flip the gate RED, proving the gate cannot bluff. (10; §1.1)
-

A

- **Access (Helix Access)** — the end-user consumer VPN app (Mullvad-style): connect, pick exit/network, toggle obfuscation, kill-switch, split-tunnel. Flutter

- UI + Rust helix-core + per-platform tunnel shim. (00 §3; 03; v04-client/helix-ui-flutter.md)
- **ACL (Access Control List)** — HelixVPN's authorization model is a Tailscale-ACL-flavored grammar (`group:X → net:Y:port/proto`) compiled to enforcement artefacts; default-deny, fail-closed. (v03-control-plane/svc-policy.md; FR-202)
 - **Advertise/route mode** — the helix-core operating mode the Connector runs in: it *advertises* served CIDRs and *routes* LAN traffic, as opposed to the Client's capture mode. (00 §3; v04-client/helix-core-rust.md; FR-703)
 - **AEAD (Authenticated Encryption with Associated Data)** — the cipher class used both by WireGuard's ChaCha20-Poly1305 and by the Shadowsocks-2022 wrap; encrypts and authenticates in one pass. (v02-data-plane/transport-shadowsocks.md; FR-009)
 - **AllowedIPs** — the WireGuard per-peer routing/cryptokey-routing list; HelixVPN's policy compiler emits a per-device AllowedIPs set as one of two enforcement artefacts (the other being the edge verdict map). An empty AllowedIPs = default-deny. (01 §8; v03-control-plane/svc-policy.md; FR-203)
 - **Anycast** — a routing technique announcing one IP from many regions (BGP) so a client reaches the nearest PoP; the D-GW-SELECT option (b) for latency-critical HA fleets. (v06-deploy/ha-and-multiregion.md; D-GW-SELECT)
 - **Anonymous device token** — an enrollment credential that requires no email/SSO/PII; a tenant mints it so a device enrolls with `users.email = NULL`, `oidc_sub = NULL`. Mullvad-parity row F15. (v05-security/identity-and-enrollment.md; v03-control-plane/svc-identity.md; FR-102; HVPN-NFR-307)
 - **Anti-bluff covenant** — the constitution's §11.4 family of rules: a PASS requires positive captured evidence that the feature works for the end user; metadata/config/absence-of-error/grep-only PASS is forbidden. The governing doctrine of Volume 8. (10; §11.4)
 - **ArkTS** — the TypeScript-based application language of HarmonyOS/OpenHarmony; the HarmonyOS shim bridges ArkTS → NAPI → the Rust `.so`. (v04-client/shim-harmonyos.md; FR-1009)
 - **Aurora OS** — the OMP-Russia Sailfish-derived mobile OS; HelixVPN targets it via the omprussia Flutter fork + a Qt/C++ tun shim + signed RPM (an enterprise SKU, Russian-hosted toolchain). (09; v04-client/shim-aurora.md; FR-1010)
 - **Audit events (audit_events)** — the durable, append-only table recording **control** actions only (who-did-what to identity/policy/devices) — never traffic/destinations/flows. Append-only enforced by REVOKE UPDATE,DELETE. (v05-security/audit-and-compliance.md; v03-control-plane/svc-telemetry.md; FR-605)
 - **Auto-ladder (transport-selection ladder)** — the client's automatic transport escalation on handshake failure: plain UDP → LWO → MASQUE/QUIC → Shadowsocks → UDP-over-TCP, driven by handshake-failure events (Mullvad's exact UX). (01 §5.2; v02-data-plane/transport-selection-ladder.md; FR-006)

B

- **Bare-link throughput** — the unobfuscated network's raw throughput, the denominator for Phase-0 gate G1 (plain-UDP WG must reach $\geq 80\%$ of it). UNVERIFIED until G1. (06; HVPN-NFR-001; G1)

- **BGP (Border Gateway Protocol)** — the inter-domain routing protocol that an Anycast HA topology (D-GW-SELECT option b) depends on. (v06-deploy/ha-and-multiregion.md)
- **boringtun** — Cloudflare's userspace WireGuard implementation (Rust); HelixVPN's helix-wg wraps it as the cross-platform userspace fallback where a kernel/native WG path is unavailable (iOS, in-process edge). (01; v02-data-plane/wireguard-core.md; D-CORE-*)
- **buf** — the Protobuf build/lint/breaking-change tool driving the helix-protoc → Go/Dart/Rust codegen; buf generate plus pinned local protoc-gen-* plugins. (v06-deploy/codegen-pipeline.md; D-CODEGEN-NET)

C

- **CA (Certificate Authority)** — the per-tenant signing authority for short-lived mTLS device certs; the tenant CA key is the single root secret to protect. MVP topology = a single online issuing CA under an offline/KMS root. (v05-security/pki-and-certs.md; v03-control-plane/svc-pki.md; FR-108; D-PKI-CA-TIER)
- **ca_chain** — the certificate chain (length ≥ 1) delivered to a device at enroll so clients pin a chain from day one; promoting CA topology grows the chain additively. (v05-security/pki-and-certs.md)
- **Capture mode** — the Client's helix-core mode: it captures the device's traffic into the tunnel (vs the Connector's advertise/route mode). (v04-client/helix-core-rust.md)
- **CarrierHealth** — the TransportConn::health() return (rtt, loss, handshakes_failed) the auto-ladder reads to decide escalation. (SPEC §7.1; v02-data-plane/transport-trait.md)
- **CGNAT** — see **100.64.0.0/10**.
- **ChaCha20-Poly1305** — WireGuard's AEAD cipher; part of the never-forked crypto core (P2). (01; v02-data-plane/wireguard-core.md; FR-001; HVPN-NFR-400)
- **challenges (vasic-digital/challenges)** — the anti-bluff Challenge-bank submodule (§11.4.27): a §11.4.169 mandatory test surface whose banks score PASS only on positive captured evidence; HelixVPN registers one Challenge per user-visible feature. (10 §5.5; v06-deploy/helix-ecosystem-integration.md; §11.4.27)
- **CIDR** — Classless Inter-Domain Routing notation for an IP prefix; a Connector advertises the CIDRs its LAN exposes (route.advertised). (v03-control-plane/svc-registry.md; FR-704)
- **Client** — the end-user role: dials in, gets an overlay IP, reaches its authorized subset of joined networks or uses the gateway as a plain privacy exit. Reuses helix-core + helix-ui. (00 §3; SPEC §3)
- **CloudNativePG (CNPG)** — a Kubernetes Postgres operator (Patroni-class) for in-cluster HA Postgres; the concrete example for D-K8S-PG-HA option (a). (v06-deploy/kubernetes.md; D-K8S-PG-HA)
- **Connect-RPC (Connect)** — the gRPC-compatible RPC framework used for the agent control channel (Coordinator.WatchNetworkMap) over QUIC/H2; browser-friendly, schema-generated. (02; v03-control-plane/svc-api.md; v03-control-plane/protobuf-spec.md)
- **CONNECT-IP** — RFC 9484 IP-proxying over HTTP; an advanced MASQUE datapath carrying raw IP packets (no inner WG) — a Phase-2 native option. (01; v02-data-plane/transport-masque-quic.md; FR-012; RFC 9484)

- **CONNECT-UDP** — RFC 9298 UDP-proxying over HTTP/3; the mechanism that wraps WireGuard datagrams to look like web traffic — the core of "MASQUE/QUIC mode". (01; v02-data-plane/transport-masque-quic.md; FR-004; RFC 9298)
- **Connector (Helix Connector)** — the network-side role/app: runs inside a private network, dials outbound to the gateway, advertises served CIDRs, routes LAN traffic for authorized clients. The in-network half of "two-way". (00 §3; SPEC §3; FR-7xx)
- **Console (Helix Console)** — the admin app: tenants, users, devices, networks, routes, policies, audit, optional billing. Flutter web+desktop, API-client only (**no helix-core**). (00 §3; v04-client/web-console.md; FR-6xx)
- **containers (vasic-digital/containers)** — the mandated container-orchestration submodule (§11.4.76/.161): every containerized HelixVPN workload (test infra, edge images) boots on-demand through its pkg/boot/pkg/compose/pkg/health primitives — rootless Podman only, no ad-hoc docker/podman commands. (05; v06-deploy/helix-ecosystem-integration.md; v06-deploy/podman-quadlets.md; §11.4.76; §11.4.161)
- **Control plane** — the Go services holding routing/policy truth (identity/registry/ipam/pki/policy/coordinator/events/telemetry/api/store); strictly separated from the data plane and never in the packet path (P1). (02; v03-control-plane/*)
- **Coordinator** — the Go control-plane brain that builds per-agent network maps from an in-memory topology graph and pushes minimal deltas with a p99 < 1s convergence SLO. (02; v03-control-plane/svc-coordinator.md; HVPN-NFR-003)
- **CoreError** — the unified 7-variant FFI error enum {NotStarted, AlreadyStarted, Config, Auth, HostFatal, BadFd, Internal} every FFI verb returns; anyhow stays orchestrator-internal and converts at the boundary. (v04-client/ffi-surface.md; REFINEMENT_NOTES R7)
- **Cover traffic** — dummy packets DAITA injects so traffic volume/timing leaks nothing to an analyst; applied above WG. (v02-data-plane/daita.md; FR-017)
- **Curve25519** — WireGuard's elliptic-curve Diffie-Hellman primitive (Noise IK); part of the never-forked crypto core. (01; FR-001)

D

- **DAITA (Defence Against AI-guided Traffic Analysis)** — Mullvad's traffic-shaping defence: constant packet sizing + cover traffic + timing perturbation applied *above* WireGuard via a maybenot state machine; orthogonal to the obfuscation transport. (01 §5.2; v02-data-plane/daita.md; FR-017; F9; D-DAITA-A)
- **Data plane** — the Rust edge + kernel/userspace WireGuard that forwards packets; keeps only counters + ephemeral routing state, no logs; fail-static when the control plane is down. (01; v02-data-plane/*; P1/P7)
- **Default-deny** — the policy posture: the empty policy denies all; a compiler error denies rather than opens (fail-closed). (v03-control-plane/svc-policy.md; v05-security/zero-trust-and-default-deny.md; FR-201)
- **Delta** — the minimal change-set a NetworkMapEvent carries for one agent (vs a full Snapshot); the unit of sub-second convergence. (SPEC §7.2; v03-control-plane/svc-coordinator.md)
- **DERP-style relay (helix-relay)** — a Tailscale-DERP-style relay the data plane falls back to when direct P2P hole-punching fails (Phase-2 NAT traversal). (01; v02-data-plane/routing-and-addressing.md; FR-020; X4)

- **Device cert** — the short-lived (≤ 24 h, auto-renew) tenant-CA-signed mTLS certificate every agent authenticates the control channel with; revocable < 1 s. (v05-security/pki-and-certs.md; FR-105/107)
- **DLQ (Dead-Letter Queue)** — the destination for poison events after a delivery-count ceiling, via Redis XAUTOCLAIM; prevents consumer spin. (v03-control-plane/svc-events.md; HVPN-NFR-207)
- **DoD** — see **8-criteria MVP DoD**.
- **DoH (DNS-over-HTTPS)** — DNS on :443; the D-KS-1 decision documents the off-tunnel-DoH split-tunnel residual rather than claiming it closed. (v05-security/kill-switch-and-dns-leak.md; D-KS-1)
- **docs_chain** — the vasic-digital/docs_chain Go engine that mechanically keeps documents + DB in sync (the §11.4.106 enforcer); HelixVPN's spec exports + workable-items sync ride it. (05; v06-deploy/helix-ecosystem-integration.md; §11.4.106)
- **DPI (Deep Packet Inspection)** — the censorship technique HelixVPN's obfuscation transports evade; the netns + nftables rig simulates a DPI UDP block for gate G2. (01; v02-data-plane/obfuscation-and-dpi.md; G2; FR-004)
- **DNS-leak protection** — forcing DNS through the tunnel and blocking plaintext off-tunnel :53, so a query never reveals destinations outside the tunnel. (v05-security/kill-switch-and-dns-leak.md; FR-503; F13)

E

- **eBPF** — the in-kernel programmable datapath HelixVPN's edge may use (alongside/with nftables) to enforce verdict maps per-peer. (01 §8; v02-data-plane/routing-and-addressing.md)
- **Edge (data-plane edge, helix-edge)** — the Rust gateway component terminating MASQUE, running the WG fast path, and applying verdict maps; consumes the same helix-transport crate as the client (byte-for-byte reuse). (01; helix_edge; D5)
- **ELD (EDID-Like Data)** — *not a HelixVPN term*; appears only in the inherited constitution's AV-testing rules and is out of scope for this product. (constitution §11.4.5)
- **Enrollment** — the device-onboarding flow: device generates its WG keypair (private key never leaves), presents an enrollment token, receives a short-lived mTLS cert + overlay IP. (v05-security/identity-and-enrollment.md; FR-102/103/104)
- **Endpoint** — the gateway's public address a transport connects to (Transport::connect(endpoint,...)); per-node GatewayInfo.endpoint is set by the coordinator. (SPEC §7.1; v03-control-plane/svc-coordinator.md)

F

- **Fail-closed** — a compiler error / missing rule denies rather than opens; the security posture of the policy compiler. (v03-control-plane/svc-policy.md; FR-201)
- **Fail-static** — the availability invariant (P1): existing tunnels keep forwarding when the control plane is down; convergence is a coordination property,

connectivity must survive its breach. (SPEC §4; v06-deploy/ha-and-multiregion.md; HVPN-NFR-200)

- **FFI (Foreign Function Interface)** — the Dart[?]Rust boundary (helix_start/stop/status_stream/...); the UI is a pure function of the FFI status stream. (SPEC §7.3; v04-client/ffi-surface.md; FR-018)
- **FIPS-203** — the NIST standard for ML-KEM (post-quantum KEM); HelixVPN's PQ handshake derives a PSK from ML-KEM mixed into classical WG. (v05-security/post-quantum.md; FR-1101; RFC/FIPS-203)
- **flutter_rust_bridge (FRB)** — the codegen tool (v2) generating the Dart[?]Rust FFI for the Flutter apps; mirrored to UniFFI for native shims. (03; v04-client/ffi-surface.md; D-FFI-*)
- **Full-tunnel** — routing *all* client traffic via the gateway exit (the plain privacy-exit / Mullvad use case), egress IP = the gateway's. (01; FR-013)
- **Fyne** — a Go desktop UI toolkit proposed by source 09_GCT and **rejected** in favour of Flutter (only Flutter reaches all 8 platforms with one codebase). (99 gap G3)

G

- **G1–G6** — the six Phase-0 spike exit gates: G1 plain-UDP throughput $\geq 80\%$, G2 MASQUE survives DPI $\geq 50\%$, G3 iOS NE memory ceiling $+\geq 30\%$ headroom (make-or-break), G4 Go-vs-Rust edge benchmark (resolves D5), G5 FRB FFI drives the core, G6 push-reconcile converges with zero polling. (SPEC §8.0; 06)
- **Gateway** — the public-VPS rendezvous role running the control plane (Go) + data-plane edge (Rust + kernel WG); accepts dials, never initiates into private networks. (00 §3; SPEC §3)
- **GatewayInfo** — the per-node coordinator-set struct (endpoint, MasqueSNI) delivered in the MapDelta; a regional failover emits gateway.failover and pushes a GatewayInfo-only delta. (v03-control-plane/svc-coordinator.md; D-GW-SELECT)
- **geoDNS** — DNS that resolves a name to the nearest region's edge IP by client geo; the recommended D-GW-SELECT option (a) for Phase-2 (no BGP dependency). (v06-deploy/ha-and-multiregion.md; D-GW-SELECT)
- **Gin** — the Go HTTP framework serving the REST (apps) + WS/SSE surface; the thin edge in front of the modular monolith. (02; v03-control-plane/svc-api.md)
- **GitOps / policy-as-code** — expressing policy as a repo that reconciles into the control plane (Phase-2). (v03-control-plane/svc-policy.md; FR-208)
- **gRPC** — the agent RPC transport (via Connect) carrying WatchNetworkMap. (02; v03-control-plane/protobuf-spec.md)

H

- **HarmonyOS NEXT** — Huawei's HarmonyOS (pure, no AOSP); HelixVPN targets it via an OpenHarmony-SIG Flutter fork + a Network Kit VPN ability (the biggest platform risk, real native tunnel-shim work). (09; v04-client/shim-harmonyos.md; FR-1009)
- **helix-core / helix_core** — the Rust data-plane core shared by client, connector, and edge: WG control, transport, reconciler, FFI. The D2 decision (Rust over Go). (03; v04-client/helix-core-rust.md; D2)

- **helix_design** — the decoupled, reusable design-system submodule (vasic-digital/helix_design) — the canonical token source + polyglot exporters; the subject of Volume 10 (OpenDesign mandate §11.4.162). (10; v10-design/*; D-DESIGN-EXTRACT; D-OD-1)
- **helix-edge / helix_edge** — see **Edge**.
- **HelixError** — the *superseded* name for the unified FFI error enum sketched in [SPEC §7.3]; the R7 reconciliation renamed it to **CoreError** (see that entry — the canonical 7-variant type every FFI verb returns). HelixError survives only as a label in the SPECIFICATION.md §7.3 FFI sketch and MUST NOT be used in new code. (SPEC §7.3; v04-client/ffi-surface.md; REFINEMENT_NOTES R7)
- **helix-ffi / helix_ffi** — the Rust crate exposing the flutter_rust_bridge + UniFFI surface to Dart/native. (03; v04-client/ffi-surface.md)
- **helix-go / helix_go** — the Go control-plane modular monolith. (02; helix_go)
- **helix-proto / helix_proto** — the Protobuf + OpenAPI schemas from which Dart/Go/Rust clients are generated (schema-first, zero drift, P8). (02; v03-control-plane/protobuf-spec.md; v06-deploy/codegen-pipeline.md)
- **HelixQA / helix_qa** — the HelixDevelopment/HelixQA anti-bluff autonomous-QA submodule + its test banks; one of the §11.4.169 mandatory test surfaces. (10 §5.6; v06-deploy/helix-ecosystem-integration.md; §11.4.27)
- **helix-relay** — see **DERP-style relay**.
- **helix-transport / helix_transport** — the Rust crate implementing the Transport trait — one obfuscation implementation, three consumers (client, connector, edge). (01 §7.1; v02-data-plane/transport-trait.md)
- **helix-ui / helix_ui** — the Flutter/Dart UI core (design system + screens) shared by all three apps via three flavors. (03; v04-client/helix-ui-flutter.md)
- **helixvpncctl** — the Cobra CLI for self-host: init / keys / enroll-token / policy / revoke; one helixvpncctl init stands up the gateway. (05; v06-deploy/helixvpncctl.md; FR-901/904)
- **Hole punching** — the NAT-traversal technique that establishes a direct P2P session by simultaneous outbound packets; relay fallback engages on failure. (v02-data-plane/routing-and-addressing.md; FR-020)
- **hostNetwork: true** — the Kubernetes pod setting that lands UDP/443 directly on the node IP (node IP = gateway IP); the edge runs as a DaemonSet with it (D-K8S-EDGE-INGRESS option a). (v06-deploy/kubernetes.md; D-K8S-EDGE-INGRESS)
- **h3** — the Rust HTTP/3 library (with quinn) implementing the MASQUE CONNECT-UDP carrier. (01; v02-data-plane/transport-masque-quic.md)
- **Hybrid PQ** — post-quantum that is layered on (never replaces) classical WG: with PQ off the tunnel is still secure; with PQ on an attacker must break both. (v05-security/post-quantum.md; FR-1102; HVPN-NFR-407)
- **Hysteria2** — a turnkey QUIC+obfuscation protocol (with the Salamander obfuscator); the D1 Camp-B primary surfaced by the plurality of analyses, kept as a quinn-tuning reference rather than the chosen primary. (01; 08; D1)

I

- **IK (Noise IK)** — the Noise-protocol handshake pattern WireGuard uses (initiator knows responder's static key); part of the never-forked crypto core. (01; v02-data-plane/wireguard-core.md; FR-001)
- **IPAM (IP Address Management)** — the control-plane service allocating from the tenant overlay pool, tracking host allocations + 4via6 mappings; collision-free under concurrent enrollment. (v03-control-plane/svc-ipam.md; FR-307)
- **iperf3** — the throughput measurement tool capturing gate G1/G2 evidence over the netns rig. (06; v08-testing/test-rig.md; G1)

J

- **JetStream (NATS JetStream)** — the persistent NATS streaming layer HelixVPN swaps to at scale (Phase-2) from MVP Redis Streams; the D3 Phase-2 transport. (02; v06-deploy/ha-and-multiregion.md; D3)
- **JNI (Java Native Interface)** — the Android bridge from the VpnService Java/Kotlin layer to the Rust .so (builder/protect/fd handoff). (v04-client/shim-android.md; FR-1005)

K

- **KEM (Key Encapsulation Mechanism)** — the public-key primitive class ML-KEM belongs to; the PQ handshake derives a shared secret/PSK from it. (v05-security/post-quantum.md; FR-1101)
- **Kill-switch** — core-owned state (driven by helix-core's state machine) that blocks all plaintext egress when the tunnel is down or escalating transports; enforced per-OS via the OS firewall. (v05-security/kill-switch-and-dns-leak.md; FR-501/502; F11)
- **KMS (Key Management Service)** — the cloud HSM-class store holding the offline CA root / backup-encryption keys; the MVP CA topology roots in KMS. (v05-security/pki-and-certs.md; 99 G1)
- **KMP (Kotlin Multiplatform)** — a client-core alternative proposed by 06_GRK, recorded as a considered-then-not-chosen option (Rust core won on the iOS NE memory ceiling). (99; 03)

L

- **last_seen_at** — the single durable presence derivative; coarsened to ≥ 5 min so it cannot reconstruct a session timeline (carries no destination). (v05-security/no-logging-as-code.md; HVPN-NFR-305)
- **LINDDUN** — a privacy threat-modeling methodology (linkability/identifiability/...); used alongside STRIDE in the threat model. (v05-security/threat-model.md)
- **LWO (Lightweight WG Obfuscation)** — a cheap keyed WG-header obfuscation + padding rung to evade naive WG-signature blocks; the second rung on the auto-ladder. (01 §5.2; v02-data-plane/transport-lwo.md; FR-005; F6)

M

- **MapResponse** — Tailscale's network-map document; HelixVPN's NetworkMapEvent (snapshot + delta) is the MapResponse-style desired-state spine. (SPEC §4; v03-control-plane/svc-coordinator.md)
- **MASQUE (Multiplexed Application Substrate over QUIC Encryption)** — the IETF working-group + mechanism (RFC 9298 CONNECT-UDP over HTTP/3) that wraps WireGuard datagrams to look like web traffic — *the* "QUIC mode". Mullvad's QUIC mode IS WG-over-MASQUE, not a separate protocol (the single most important architectural correction). (01; v02-data-plane/transport-masque-quic.md; D1; FR-004; RFC 9298)
- **maybenot** — the Rust traffic-shaping state-machine framework that implements DAITA's padding/timing defences. (v02-data-plane/daita.md; FR-017)
- **Melos** — the Dart/Flutter monorepo tool managing the helix-ui multi-package tree (3 flavors). (03; v04-client/helix-ui-flutter.md)
- **ML-KEM (Module-Lattice KEM)** — the NIST FIPS-203 post-quantum KEM HelixVPN uses to derive a PSK mixed into the WG handshake (hybrid, never PQ-only). Exact byte sizes UNVERIFIED pending the PQ doc's cited confirmation. (v05-security/post-quantum.md; FR-1101; FIPS-203)
- **Modular monolith** — the Go control-plane architecture: many in-process domain modules (identity/registry/ipam/pki/policy/coordinator/events/telemetry) behind one deployable, package-boundary-disciplined (no cross-store import). (02; v03-control-plane/architecture-and-wiring.md)
- **mTLS (mutual TLS)** — both sides present certs; HelixVPN authenticates the control channel with short-lived tenant-CA-signed mTLS device certs (control auth shares no key with the WG data channel). (v05-security/pki-and-certs.md; FR-105/106)
- **MTU (Maximum Transmission Unit)** — the path packet-size limit; WG MTU 1420 on plain UDP, reduced under MASQUE, so WG-in-transport never fragments. (v02-data-plane/transport-plain-udp.md; FR-016; HVPN-NFR-008)
- **Multi-hop** — entry/exit separation via nested WireGuard with per-hop keys (entry and exit can be in different jurisdictions); Phase-2. (v02-data-plane/multihop.md; FR-401; F10)
- **Multi-network model** — the two-way reverse-tunnel topology where one user reaches N joined private networks via per-user ACL routing. (00 §4.2; X1)

N

- **NAPI (Native API)** — the OpenHarmony native-binding layer the HarmonyOS shim uses (ArkTS → NAPI → .so). (v04-client/shim-harmonyos.md)
- **NAT traversal** — establishing connectivity across NATs (STUN-like discovery + hole punching + relay fallback); Phase-2 direct-P2P. (v02-data-plane/routing-and-addressing.md; FR-020; X4)
- **NATS / NATS JetStream** — the message system HelixVPN scales the event bus to (Phase-2) from MVP Redis Streams; see **JetStream**, **D3**. (02; D3)
- **NEPacketTunnelProvider (NE)** — Apple's Network Extension packet-tunnel provider class (iOS/macOS); runs the Rust core under a hard memory ceiling — gate G3 (make-or-break, the reason the core is Rust). (v04-client/shim-apple.md; FR-1004; G3; HVPN-NFR-500)
- **Network Kit** — HarmonyOS's networking ability set providing the VPN-tunnel capability the HarmonyOS shim drives. (v04-client/shim-harmonyos.md)

- **Network map** — the per-agent desired-state document (overlay IP, *policy-filtered* peers, routes, transport policy, DNS, kill-switch posture) pushed over WatchNetworkMap. Peers are need-to-know-filtered server-side. (SPEC §7.2; v03-control-plane/svc-coordinator.md; FR-207)
- **NetworkMapEvent** — the streamed protobuf message carrying either a Snapshot or a Delta plus a monotonic version. (SPEC §7.2; v03-control-plane/protobuf-spec.md)
- **nftables** — the Linux packet-filter framework HelixVPN's edge compiles verdict maps into (and the rig uses to simulate a DPI UDP block). (01 §8; v02-data-plane/routing-and-addressing.md)
- **Noise** — the cryptographic protocol framework WireGuard's IK handshake is built on. (01; FR-001)
- **No-logging-as-code** — the architectural guarantee that no durable connection/traffic table exists; a CI schema-lint fails the build if one appears, and the lint is asserted against the *deployed* DB (runtime signature, §11.4.108). (v05-security/no-logging-as-code.md; FR-801; HVPN-NFR-300/308)

O

- **Obfuscation** — making the encrypted WG datagrams look like something else on the wire; a *pluggable layer beneath* WG (the Transport trait), never a crypto fork. (01 §5.2; v02-data-plane/obfuscation-and-dpi.md)
- **OIDC (OpenID Connect)** — the identity protocol for human principals (tenant owner/admin/member); coexists with anonymous device tokens. (v03-control-plane/svc-identity.md; FR-101)
- **OpenAPI** — the REST contract spec (spec-first, hand-authored) from which Dart/TS app clients generate; handlers validated against it. (v06-deploy/codegen-pipeline.md; D-OPENAPI-AUTHORING)
- **OpenDesign** — the mandatory (§11.4.162) design authoring/refinement system; in HelixVPN it is the design *authoring* layer while *helix_design* owns the canonical token source + polyglot exporters (the D-OD-1 reconciliation). (10; v10-design/opendesign-foundation.md; D-OD-1)
- **OpenHarmony** — the open-source base of HarmonyOS; its SIG maintains the Flutter fork the HarmonyOS shim builds on. (v04-client/shim-harmonyos.md)
- **Overlay IP** — the stable per-node address in the tenant overlay (ULA IPv6 /48 per tenant; advertised IPv4 LANs mapped via 4via6); persists across reconnects. (v03-control-plane/svc-ipam.md; FR-304)

P

- **Patroni** — a Postgres HA template (leader election + streaming replication); the HA Postgres story for multi-region (D-K8S-PG-HA option a / CloudNativePG). (v06-deploy/ha-and-multiregion.md; v06-deploy/kubernetes.md)
- **PDP / PEP (Policy Decision/Enforcement Point)** — the zero-trust split: the control plane *decides* (compiles policy) at compile-time; the edge *enforces* (AllowedIPs + verdict maps) at runtime. (v05-security/zero-trust-and-default-deny.md)
- **Podman (rootless)** — the mandated container runtime (§11.4.161); one rootless pod for a homelab, the same images scaling to HA. Docker rootful / sudo forbidden. (05; v06-deploy/podman-quadlets.md; FR-902; §11.4.161)

- **Port-hopping** — periodically changing the WG/transport port (and disguising as :443/:53) to evade port-based blocks. (v02-data-plane/transport-masque-quic.md; FR-008; F8)
- **Post-quantum (PQ)** — see **Hybrid PQ, ML-KEM, Rosenpass**.
- **Presence** — live, ephemeral reachability state kept in TTL'd Redis (never a durable table); loss of Redis loses no durable state (fail-static). (v03-control-plane/svc-events.md; HVPN-NFR-304)
- **Protobuf (Protocol Buffers)** — the agent-contract schema language (helix.coordinator.v1); Dart/Go/Rust clients generate from it. (02; v03-control-plane/protobuf-spec.md; P8)
- **PSK (Pre-Shared Key)** — WireGuard's optional symmetric key slot; HelixVPN mixes the ML-KEM-derived secret into it for hybrid PQ. (v05-security/post-quantum.md; FR-1101)
- **PWU (Parallel Work Unit)** — a self-contained workable item in the §11.4.58/.93 parallel execution model. (SPEC §11; §11.4.58)

Q

- **quadlet** — a systemd unit format (Podman) declaring rootless containers/pods; HelixVPN ships the gateway as quadlet units (NET_ADMIN, :443/udp, one pod, read-only rootfs). (v06-deploy/podman-quadlets.md; FR-906)
- **QUIC** — the UDP-based transport protocol underlying HTTP/3 and MASQUE; HelixVPN's "QUIC mode" is WG-over-MASQUE-over-QUIC, not a bespoke QUIC VPN. (01; v02-data-plane/transport-masque-quic.md; D1)
- **quinn** — the Rust QUIC implementation (with h3) HelixVPN uses for the MASQUE carrier and as a Hysteria2-tuning reference. (01; v02-data-plane/transport-masque-quic.md)

R

- **Reconciler** — the helix-core loop that drives the local state to the pushed network map (desired-state, no polling); reconnection state machine lives here. (v02-data-plane/orchestrator-and-state.md; FR-015; G6)
- **Redis / Redis Streams** — the MVP event bus + ephemeral KV (TTL presence); consumer groups + XAUTHCLAIM DLQ; swapped to NATS at scale (D3). (02; v03-control-plane/svc-events.md; D3)
- **Reverse tunnel** — the founding topology: internal hosts dial *outbound* to a public gateway, which relays/routes — no inbound port-forward on any private network. (00; SPEC §1; X2)
- **RFC 2119** — the standard defining MUST/SHOULD/MAY force, used throughout the spec. (SPEC §0)
- **RLS (Row-Level Security)** — PostgreSQL's per-row tenant isolation (FORCE ROW LEVEL SECURITY), enforcing multi-tenancy at the database, not only the app layer. (v03-control-plane/data-model-ddl.md; FR-110; HVPN-NFR-408)
- **Rosenpass** — a post-quantum WG key-exchange add-on evaluated as an alternative to the ML-KEM-PSK approach; its Kyber-vs-ML-KEM status is UNVERIFIED pending the PQ doc's cited confirmation. (v05-security/post-quantum.md; FR-1103)

- **Riverpod** — the Flutter state-management library; UI state is a pure function of the FFI status stream + Console WS/SSE folding. (03; v04-client/state-management.md)
- **RPM** — the package format for the signed Aurora OS build. (v04-client/shim-aurora.md; FR-1010)
- **RTO / RPO (Recovery Time/Point Objective)** — the DR budgets owned by v06-deploy/disaster-recovery.md (closing ledger gap G1); NFR-205 names the owner, not a guessed number. (HVPN-NFR-205; 99 G1)
- **Runtime signature** — the §11.4.108 definition-of-done: a fix is done only when its one machine-checkable observable verifies on a *clean* deployment (e.g. schema-lint green against the deployed DB). (10; §11.4.108; HVPN-NFR-308)
- **runHelixApp(flavor, home, capabilities)** — the single Flutter entrypoint building all three flavors (Access/Connector/Console) from one tree with per-flavor capability gating. (v04-client/helix-ui-flutter.md; FR-1001; HVPN-NFR-600)

S

- **S0–S8** — the Phase-0 spike milestones (the gate-bearing work-breakdown steps). (06)
- **Salamander** — Hysteria2's obfuscation layer (the D1 Camp-B knob set kept as a tuning reference). (01; D1)
- **Shadowsocks (Shadowsocks-2022)** — an AEAD proxy protocol; HelixVPN wraps WG in it as an obfuscation rung for QUIC/UDP-hostile networks (via shadowsocks-rust). (01; v02-data-plane/transport-shadowsocks.md; FR-009; F4)
- **Shim (tunnel shim / TunnelPlatform)** — the thin per-platform native layer attaching the OS tun device to the Rust core (Apple NE, Android VpnService+JNI, Windows wireguard-nt+service, Linux, HarmonyOS, Aurora; Web = none). (03; v04-client/shim-*.md)
- **sing-box** — a Go universal transport-multiplexer framework proposed by 06_GRK and **rejected** (Go conflicts with the Rust-core D2 + iOS memory ceiling; custom Rust helix-transport gives byte-for-byte client[?]edge reuse). (99 gap G2)
- **Snapshot** — the full desired-state a NetworkMapEvent carries when an agent opens the stream with known_version=0 (vs a Delta). (SPEC §7.2; v03-control-plane/svc-coordinator.md)
- **SPIFFE** — the secure-production-identity framework; HelixVPN's mTLS device cert SAN is a SPIFFE-style ID binding it to a device_id. (v05-security/pki-and-certs.md)
- **Split horizon** — the default that Connectors can't reach each other and Clients can't reach un-granted networks absent an explicit rule. (v03-control-plane/svc-policy.md; FR-206)
- **Split tunneling** — routing some traffic in-tunnel and some out (per-route, and per-app on Android/desktop). (01; FR-014; F12)
- **SSE (Server-Sent Events)** — a live one-way push channel the Console may use (alongside WS) for live state. (v03-control-plane/svc-api.md; FR-603)
- **STRIDE** — the threat-modeling methodology (spoofing/tampering/...); paired with LINDDUN in the threat model. (v05-security/threat-model.md)
- **STUN** — the NAT-discovery mechanism the P2P path uses to learn external mappings before hole punching. (v02-data-plane/routing-and-addressing.md; FR-020)

T

- **Tailscale** — the prior-art coordination model HelixVPN borrows from (network-map / MapResponse, 4via6, ACL grammar); HelixVPN is the self-hosted Mullvad-grade union. (00; SPEC §1)
- **Telemetry** — the counts-only health/audit service (Prometheus metrics, audit_events); carries no flows/destinations; label set $\subseteq$ allow-list (no tenant_id/device_id/*_ip). (v03-control-plane/svc-telemetry.md; FR-803; HVPN-NFR-306)
- **Terraform** — the IaC tool for the multi-region/DR provisioning + region-failover runbook (Phase-2/DR). (v06-deploy/disaster-recovery.md; 99 G1)
- **Transport trait** — the single Rust abstraction beneath WireGuard: connect/accept yield a TransportConn (send/recv WG datagrams + health). One implementation, three consumers. (SPEC §7.1; v02-data-plane/transport-trait.md; FR-003)
- **TransportConn** — a live carrier passing WG datagrams in/out plus the CarrierHealth the auto-ladder reads. (SPEC §7.1; v02-data-plane/transport-trait.md)
- **TransportKind** — the stable transport identifier enum (PlainUdp | MasqueH3 | Shadowsocks | UdpOverTcp | Lwo) used by the ladder + counts-only telemetry. (SPEC §7.1)
- **tc netem** — the Linux traffic-control network-emulation tool injecting loss/jitter into the Phase-0 netns rig. (06; v08-testing/test-rig.md)
- **tun / TUN device** — the OS virtual network interface the tunnel rides; absent in a browser (the reason Web is Console-only, not a system VPN). (03; v04-client/web-console.md; HVPN-NFR-607)
- **TunnelStatus / FfiTunnelStatus** — the FFI status enum the UI subscribes to (Disconnected | Connecting(transport) | Connected | Reconnecting | Failed); FfiTunnelStatus is the actual Rust identifier, ffi::TunnelStatus the path-qualified name of the same type. (SPEC §7.3; v04-client/ffi-surface.md; FR-018; REFINEMENT_NOTES R7)
- **Two-way model** — the gateway stitching the network-side leg (Connector→Gateway) and user-side leg (Client→Gateway) without either side opening an inbound port. (00 §4; FR-306; X2)

U

- **UDP-over-TCP (UoT / udp2tcp)** — the last-resort transport tunnelling UDP inside TCP when UDP is fully blocked. (01; v02-data-plane/transport-udp-over-tcp.md; FR-010; F5)
- **ULA (Unique Local Address)** — IPv6 private space (fc00::/7); HelixVPN assigns each tenant a ULA /48 as the overlay, into which advertised IPv4 LANs are mapped via 4via6. (v02-data-plane/routing-and-addressing.md; v03-control-plane/svc-ipam.md; FR-303; D4)
- **UniFFI** — Mozilla's multi-language FFI binding generator; HelixVPN mirrors the flutter_rust_bridge surface to UniFFI for the native (Swift/Kotlin) shims. (03; v04-client/ffi-surface.md)

V

- **Verdict map** — the nftables/eBPF map compiled from policy that grants/denies per-peer reachability at the edge, keyed (`src_overlay_ip`, `dst_cidr`, `l4proto`, `dport`). The second policy enforcement artefact (with AllowedIPs). (01 §8; `v03-control-plane/svc-policy.md`; FR-203)
- **version (network-map version)** — the monotonic counter on each `NetworkMapEvent`; agents resume a stream by `known_version`. (SPEC §7.2; HVPN-NFR-204)
- **vision_engine** — the vasic-digital video-evidence/QA submodule (OCR/vision per §11.4.107/.158) wired for recorded-evidence test verdicts. (10 §5.14; `v06-deploy/helix-ecosystem-integration.md`)
- **VpnService** — Android's VPN API; the Android shim integrates it (builder/protect/fd handoff) with the Rust core via JNI, with background-kill resilience. (`v04-client/shim-android.md`; FR-1005)
- **VMAF / SSIM / ΔE2000** — full-reference video-quality metrics from the inherited constitution's AV-testing rules; **out of scope** for HelixVPN (no media-playback surface). (constitution §11.4.107)

W

- **WASM MASQUE proxy** — an optional in-page WebAssembly proxy that proxies the *browser's own* traffic to a joined network — never a system VPN (a browser has no TUN). Phase-3, honestly scoped. (09; `v04-client/web-console.md`; FR-1011; HVPN-NFR-607)
- **WatchNetworkMap** — the server-streaming RPC every agent opens once: a full Snapshot then a Delta stream of policy-filtered desired state; replaces all polling ($p99 < 1s$ convergence). (SPEC §7.2; `v03-control-plane/svc-coordinator.md`; `v03-control-plane/protobuf-spec.md`; FR-205/207)
- **WFP (Windows Filtering Platform)** — the Windows kernel filter API the Windows shim uses for split-tunnel + kill-switch. (`v04-client/shim-windows.md`; FR-1007)
- **WireGuard (WG)** — the cryptographic core everywhere (Noise IK, Curve25519, ChaCha20-Poly1305); never forked, obfuscation is layered beneath it. (01; `v02-data-plane/wireguard-core.md`; FR-001; P2)
- **wireguard-nt / wintun** — the Windows kernel WireGuard / TUN drivers the Windows client integrates behind a privileged service with named-pipe IPC. (`v04-client/shim-windows.md`; FR-1007)
- **WS (WebSocket)** — the bidirectional live channel (with SSE) the Console/clients use for live state. (`v03-control-plane/svc-api.md`; FR-603)

Z

- **Zero-trust** — the security posture: default-deny + need-to-know peer filtering + PDP/PEP split; a node only learns peers a compiled rule grants. (`v05-security/zero-trust-and-default-deny.md`; FR-201/207; HVPN-NFR-402)
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RFC index

The protocol RFCs the specification binds to. Numbers cited are from the source research; the *interop* claim each one underpins is the owning doc's gate to prove (marked UNVERIFIED where a measured/interop result is involved).

RFC	Title (as used here)	Role in HelixVPN	Owning doc
RFC 2119	Key words for requirement levels (MUST/SHOULD/MAY)	Requirement-force convention across the whole spec	SPEC §0
RFC 9298	Proxying UDP in HTTP (CONNECT-UDP over HTTP/3)	The MASQUE mechanism wrapping WG datagrams — "QUIC mode"	v02-data-plane/transport-masque-quic.md
RFC 9297	HTTP Datagrams and the Capsule Protocol	The datagram framing MASQUE CONNECT-UDP rides	v02-data-plane/transport-masque-quic.md
RFC 9221	Unreliable Datagram Extension to QUIC	QUIC DATAGRAM frames carrying the WG datagrams (low-overhead path)	v02-data-plane/transport-masque-quic.md
RFC 9484	Proxying IP in HTTP (CONNECT-IP)	Advanced native IP-over-HTTP/3 datapath (no inner WG), Phase-2	v02-data-plane/transport-masque-quic.md; FR-012
FIPS-203	Module-Lattice-Based KEM (ML-KEM)	The PQ KEM deriving the hybrid PSK	v05-security/post-quantum.md; FR-1101

UNVERIFIED note (§11.4.6). The RFC *numbers and titles* are as cited in the source research and the owning docs; where the owning doc itself flags an external fact pending the re-run web-research pass (e.g. exact ML-KEM byte sizes, the WireGuard whitepaper PDF), that flag carries here. The deep-research appendix ([../11-deep-research-appendix.md](#)) holds the cited (URL + access-date) corpus for 10/10 angles as of Revision 3.

Sources verified

- [../SPECIFICATION.md](#) §3 (roles), §4 (principles), §5 (architecture + transport layering), §7 (the three cross-cutting contracts: Transport trait, WatchNetworkMap, FFI), §9 (decision register D1–D8), §11 (the 20-row spine glossary this document expands).
- [../MASTER_INDEX.md](#) Volumes 0–10 (owning-doc filenames for every term).
- [../v01-product/functional-requirements.md](#) and [nonfunctional-requirements.md](#) (FR/NFR ids cross-referenced per term).
- The volume nano-detail docs cited inline: v02-data-plane/* (transports, WG, routing, DAITA, ladder), v03-control-plane/* (coordinator, events, ipam, pki, policy, telemetry, ddl, protobuf), v04-client/* (ffi-surface, helix-core-

- rust, shims, web-console), v05-security/* (threat-model, pki-and-certs, no-logging, kill-switch, post-quantum, audit), v06-deploy/* (codegen, podman-quadlets, kubernetes, ha-and-multiregion, helixvpncntl, helix-ecosystem-integration), v10-design/* (opendesign-foundation, design-tokens).
- ../99-source-coverage-ledger.md (the rejected-alternative terms: sing-box G2, Fyne G3, KMP, CGNAT/100.64).

Constitution bindings: §11.4.44 (revision header), §11.4.6 (no-guessing — UNVERIFIED on terms whose numeric/interop claim awaits a named gate; no term invented), §11.4.65/.153 (HTML+PDF[+DOCX] exports follow in refinement), §11.4.35 (this glossary is the consumer-side reference; the spine glossary §11 is canonical for the 20 spine terms).

Honesty note (§11.4.6): every entry traces to a cited source doc or RFC; no definition is asserted from memory. Terms inherited from the constitution but out of HelixVPN scope (ELD, VMAF/SSIM/ΔE2000) are marked as such rather than silently omitted (§11.4.118).